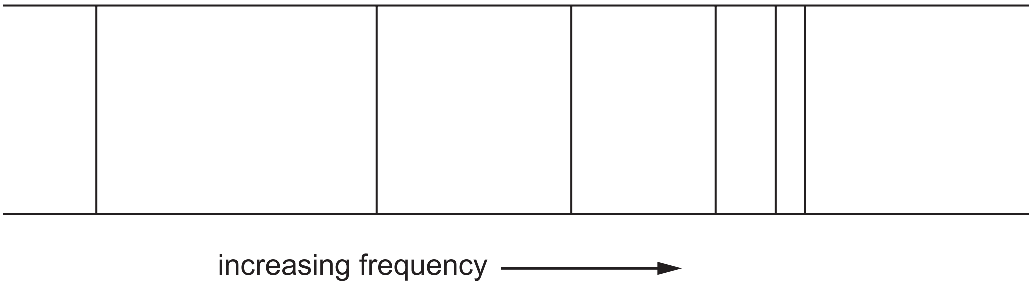
Examiner only

8. *(a)* The diagram below shows part of the atomic emission spectrum of hydrogen.



(i) Use the letter **A** to label the line of longest wavelength on the diagram. [1]

(ii) Explain why hydrogen atoms emit only certain definite frequencies of visible light. [2]

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(b) The ionisation energy of a hydrogen atom is 2.18×10^{-21} kJ.

(i) Explain what this statement means. [2]

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(ii) Calculate the minimum frequency of radiation required to ionise a hydrogen atom in its ground state. [3]

Frequency = ... s^{-1}

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(c) Hydrazine is a compound of hydrogen and nitrogen only. It is a colourless, flammable liquid which was used in various rocket fuels.

0.160 g of hydrazine on vaporisation at 398 K and 1 atm pressure has a volume of 163 cm^3.

Calculate its volume at 273 K and 1 atm pressure and hence show that its molecular formula is N_2H_4. [3]

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(d) (i) Draw a dot and cross diagram to show the electron arrangement in hydrazine, N_2H_4. Show outer electrons only. [2]

(ii) Hydrazine contains polar covalent bonds between nitrogen and hydrogen atoms. State what is meant by a *polar* covalent bond. [1]

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(e) Hydrazine acts as a base in a similar way to ammonia.

Suggest an equation for the equilibrium formed when hydrazine dissolves in water. [1]

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